

# Economics — A Four-Week Course

*Every lecture (short version) plus sample homework and exam problems with an answer key.*

## Week 1 — Foundations: scarcity, trade, and markets

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What economics is and why it exists; the production possibilities frontier; comparative advantage and trade; supply and demand; market equilibrium; elasticity; price controls and taxes.

### 1.1 What economics is and why it exists

*Economics exists because wants are unlimited and resources are not.*

#### What economics is and why it exists

Economics is the study of how people and societies **allocate scarce resources among competing uses**. Labor, land, capital, and time are finite; wants are not. Every decision to use a resource one way is therefore a decision not to use it another way.

#### Opportunity cost

The true cost of any choice is its **opportunity cost**: the value of the best alternative given up. Money prices are only part of it — time, effort, and forgone income count too. Rational choice means acting whenever the *marginal* benefit of one more unit exceeds its marginal cost.

#### The three core questions

Every economic system must answer three questions:

- **What** to produce — which goods and services, in what quantities?
- **How** to produce it — with which combination of labor, capital, and technology?
- **For whom** — how is the output distributed among people?

**Microeconomics** studies these questions for individual households, firms, and markets; **macroeconomics** studies the economy as a whole — output, inflation, unemployment, and growth.

#### A real example

A student who spends four years in college pays tuition — say \$15,000 a year — but also gives up the roughly \$30,000 a year she could have earned working. The opportunity cost of college is about \$45,000 a year, and the largest component never appears on any bill. This is why college is costliest for exactly the people with the best outside options, and why a star athlete may rationally leave school early.

## 1.2 The production possibilities frontier

*The menu of what an economy can produce, and the cost of moving along it.*

### The production possibilities frontier

The **production possibilities frontier (PPF)** shows the maximum combinations of two goods an economy can produce with its given resources and technology. Points *on* the frontier are efficient, points *inside* it waste resources, and points *outside* it are unattainable.

### The slope is opportunity cost

Moving along the frontier means producing more of one good only by producing less of the other:

$$\text{Opportunity cost of } X = -\Delta Y / \Delta X.$$

The frontier is usually **bowed outward** because resources are specialized: the first workers moved from farming to factories are the worst farmers, but later ones are the best, so each extra unit of manufactured goods costs more food than the last. This is the law of **increasing opportunity cost**.

### Shifting the frontier

The PPF shifts outward — **economic growth** — when the economy acquires more resources (labor, capital) or better technology. Choosing more capital goods today at the expense of consumption shifts tomorrow's frontier further out.

### A real example

When the United States entered World War II, auto plants were converted to build tanks and bombers, and civilian car production essentially stopped from 1942 to 1945 — a movement along the frontier from butter to guns. But because the Depression had left the economy far *inside* its frontier in 1941, the country initially increased both military output and civilian consumption at once, by putting idle workers and factories back to work.

## 1.3 Comparative advantage and trade

*Trade pays whenever opportunity costs differ — even if one side is better at everything.*

### Comparative advantage and trade

A producer has an **absolute advantage** if it can make a good with fewer resources. It has a **comparative advantage** if it can make the good at a lower *opportunity cost*. David Ricardo showed in 1817 that comparative advantage, not absolute advantage, determines who should produce what.

### The rule

With unit labor requirements  $a$  (hours per unit), country A's opportunity cost of good X in terms of Y is

$$OC_X^A = a_X^A / a_Y^A.$$

Each country should specialize in the good for which its opportunity cost is lower and trade for the other. Both gain as long as the **terms of trade** lie between the two countries'

opportunity costs.

### Winners and losers

Trade raises total output and consumption in every participating country, but not every person gains: workers and firms in import-competing industries lose. That distribution of gains is why trade is economically sound in aggregate yet politically contested.

### A real example

In Ricardo's own example, Portugal needs 90 hours of labor for a unit of cloth and 80 for a unit of wine; England needs 100 and 120. Portugal is better at both. Yet Portugal's opportunity cost of wine is  $80/90 \approx 0.89$  units of cloth, while England's is  $120/100 = 1.2$ . Portugal specializes in wine, England in cloth, and after trading both countries consume more of both goods than they could alone.

## 1.4 Supply and demand

*Two curves that summarize every buyer and every seller in a market.*

### Supply and demand

The **demand curve** shows how much buyers are willing to purchase at each price; by the **law of demand** it slopes downward. The **supply curve** shows how much sellers are willing to offer at each price; it slopes upward because higher prices cover the higher marginal cost of extra output.

### Linear forms

The simplest model uses straight lines:

$$Q_d = a - bP, Q_s = c + dP,$$

where  $b$  and  $d$  measure how responsive buyers and sellers are to price.

### Movements versus shifts

A change in the good's *own price* moves along a curve. A change in anything else *shifts* the whole curve:

- **Demand shifters** — income, prices of substitutes and complements, tastes, expectations, number of buyers.
- **Supply shifters** — input prices, technology, expectations, number of sellers, taxes and subsidies.

Confusing the two — a "change in quantity demanded" versus a "change in demand" — is the most common error in the subject.

### A real example

In spring 2020, lockdowns sent millions of people into their kitchens, and demand for flour and baking yeast shifted sharply right; supermarket shelves emptied within weeks. At the same moment, demand for jet fuel collapsed as airlines grounded fleets. Neither change involved an initial change in the goods' own prices — both were shifts of the entire demand curve.

## 1.5 Market equilibrium

*Where the curves cross, and how markets get there.*

### Market equilibrium

A market is in **equilibrium** at the price where the quantity buyers want equals the quantity sellers offer. At that price there is no pressure for change.

#### The equilibrium condition

$$Q_d(P^*) = Q_s(P^*).$$

With linear curves, setting  $a - bP = c + dP$  gives

$$P^* = (a - c) / (b + d).$$

Below  $P^*$  there is a **shortage**: frustrated buyers bid the price up. Above  $P^*$  there is a **surplus**: unsold inventory forces the price down.

#### Comparative statics

Shifting one curve and comparing equilibria gives predictions: demand rising raises both price and quantity; supply rising lowers price and raises quantity. When both curves shift, the direction of one variable is ambiguous and depends on which shift is larger.

#### A real example

With  $Q_d = 100 - 2P$  and  $Q_s = 20 + 2P$ , equilibrium is  $P^* = 80/4 = 20$  and  $Q^* = 60$ . In the real world, the 2021–22 semiconductor shortage cut the supply of new cars; buyers spilled into the used-car market, demand there shifted right, and U.S. used-car prices rose by roughly 40 percent in about a year — exactly the comparative-statics prediction.

## 1.6 Elasticity

*How strongly quantity responds to price.*

### Elasticity

Knowing that demand slopes down is not enough; we need to know *how much* quantity changes when price changes. That sensitivity is **elasticity**.

#### Price elasticity of demand

$$E_d = \% \Delta Q_d / \% \Delta P.$$

Because percentage changes depend on the starting point, economists use the **midpoint formula**:

$$E = [(Q_2 - Q_1) / ((Q_1 + Q_2)/2)] / [(P_2 - P_1) / ((P_1 + P_2)/2)].$$

Demand is **elastic** if  $|E| > 1$ , **inelastic** if  $|E| < 1$ , and **unit elastic** if  $|E| = 1$ .

#### Determinants and revenue

- Close substitutes make demand more elastic.
- Necessities are inelastic; luxuries are elastic.
- Goods that take a large share of the budget are more elastic.

- Demand is more elastic over longer time horizons.

Elasticity decides what a price change does to revenue: raising price raises total revenue when demand is inelastic and lowers it when demand is elastic.

### A real example

Cigarette demand has an elasticity of roughly  $-0.4$ : a 10 percent tax-driven price increase cuts consumption by about 4 percent. Because demand is inelastic, cigarette taxes raise substantial revenue *and* reduce smoking. Teenagers, with less income and less entrenched habits, respond more strongly — which is why public-health economists favor tobacco taxes as youth policy.

## 1.7 Price controls and taxes

*What happens when government overrides the market price.*

### Price controls and taxes

A **price ceiling** set below equilibrium (rent control) creates a shortage; a **price floor** set above equilibrium (a minimum wage, agricultural supports) creates a surplus. In both cases the market no longer clears, and non-price rationing — queues, discrimination, lower quality — takes over.

### Taxes and incidence

A per-unit tax  $t$  drives a wedge between what buyers pay and sellers receive:

$$P_b - P_s = t.$$

Who bears the tax depends on elasticities, not on who legally writes the check. The buyers' share is

$$\text{Buyers' share} = E_s / (E_s + |E_d|).$$

The more inelastic side of the market bears more of the burden.

### Deadweight loss

Because the tax reduces the quantity traded, some mutually beneficial trades vanish. The lost surplus is the **deadweight loss**:

$$DWL = \frac{1}{2} \cdot t \cdot \Delta Q.$$

### A real example

San Francisco extended rent control to many older buildings in 1994. A 2019 Stanford study found that affected landlords reduced the supply of rental housing they offered by about 15 percent — converting units to condos and owner-occupied housing. Rent control helped the tenants who kept their apartments but shrank the rental stock and pushed up rents for everyone else: the shortage the ceiling model predicts.

## Week 2 — Consumers, firms, and market structure

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Consumer choice and utility; consumer and producer surplus; production and costs; perfect competition; monopoly; oligopoly and game theory; externalities and public goods.

### 2.1 Consumer choice and utility

*How a budget-constrained consumer picks the best bundle.*

#### Consumer choice and utility

Consumers choose among bundles of goods to maximize **utility** — their satisfaction — subject to a **budget constraint**. For two goods X and Y with income I:

$$P_x X + P_y Y = I.$$

#### The optimum

The best bundle equalizes the marginal utility per dollar across goods:

$$MU_x / P_x = MU_y / P_y.$$

Equivalently, the consumer's **marginal rate of substitution** equals the price ratio:  $MRS = P_x / P_y$ . If the last dollar spent on X yields more utility than the last dollar on Y, shifting spending toward X makes the consumer better off.

#### Diminishing marginal utility

Each additional unit of a good adds less satisfaction than the one before. This is what makes demand slope downward, and it decomposes any price change into a **substitution effect** (the good is now relatively cheaper) and an **income effect** (the consumer is now effectively richer).

#### A real example

Adam Smith's **water-diamond paradox** asked why water, essential to life, is cheap while diamonds, useless for survival, are expensive. Marginal utility resolves it: water is so abundant that the utility of the *last* gallon is tiny, while diamonds are so scarce that the last one is highly valued. Prices track marginal utility, not total utility.

### 2.2 Consumer and producer surplus

*Measuring the gains from trade in dollars.*

#### Consumer and producer surplus

**Consumer surplus** is the difference between what buyers are willing to pay and what they actually pay. **Producer surplus** is the difference between the price sellers receive and the minimum they would accept (their marginal cost). Together they measure the total value a market creates.

## Measuring surplus

With linear curves, each surplus is a triangle:

$$CS = \frac{1}{2} (P_{\max} - P^*) Q^*, PS = \frac{1}{2} (P^* - P_{\min}) Q^*,$$

where  $P_{\max}$  is the demand curve's choke price and  $P_{\min}$  the supply curve's intercept. **Total surplus** = CS + PS.

## Efficiency

The **first welfare theorem** states that a competitive equilibrium maximizes total surplus: every trade worth more to a buyer than it costs a seller happens, and no trade that destroys value does. Any policy that moves quantity away from  $Q^*$  — a tax, a quota, a price control — shrinks the total.

## A real example

A concert has a demand choke price of \$100, an equilibrium ticket price of \$40, and 3,000 tickets sold. Consumer surplus is  $\frac{1}{2}(60)(3,000) = \$90,000$ . If the lowest price at which the promoter would supply a seat is \$10, producer surplus is  $\frac{1}{2}(30)(3,000) = \$45,000$ . The \$135,000 total is the value the concert creates beyond its cost — the pie that ticket scalpers and price caps fight over.

## 2.3 Production and costs

*From inputs to output, and what each unit costs.*

### Production and costs

A firm transforms inputs into output through a **production function**  $Q = f(K, L)$ . In the **short run** some inputs (plant, equipment) are fixed; in the **long run** all inputs are variable. With capital fixed, adding workers eventually yields **diminishing marginal returns**.

### Cost concepts

$$TC = FC + VC, ATC = TC / Q, MC = \Delta TC / \Delta Q.$$

Fixed costs do not vary with output; variable costs do. **Marginal cost** — the cost of one more unit — is the number that drives every production decision. MC cuts ATC at ATC's minimum: when the marginal unit costs less than the average, it pulls the average down.

### Economies of scale

In the long run, **average cost** typically falls as the firm grows (economies of scale — specialization, spreading fixed costs), then flattens, and may eventually rise (diseconomies of scale — coordination problems). The shape of this curve determines how many firms an industry can support.

### A real example

A leading-edge semiconductor fabrication plant costs more than \$20 billion to build, but the marginal cost of producing one more chip in an existing fab is small. Average cost therefore plunges with volume, so only a handful of firms worldwide can operate at efficient scale — economies of scale explain why chip manufacturing is so concentrated.

## 2.4 Perfect competition

*Many firms, identical products, and price equal to marginal cost.*

### Perfect competition

A perfectly competitive market has many small firms, a homogeneous product, free entry and exit, and full information. Each firm is a **price taker**: it can sell all it wants at the market price and nothing above it, so its marginal revenue equals the price.

#### Profit maximization

$$\pi = TR - TC, \text{ produce where } P = MR = MC.$$

In the short run the firm operates as long as price covers **average variable cost**; if  $P < AVC$  it shuts down, because producing loses more than its fixed costs.

#### The long run

Economic profits attract entry, which shifts supply right and pushes price down; losses cause exit. The process stops only when economic profit is zero:

$$P = MC = \min ATC.$$

Firms produce at the lowest possible cost, and price equals marginal cost — the market is both productively and allocatively efficient.

#### A real example

U.S. corn farming is close to perfect competition: hundreds of thousands of farms, an undifferentiated product, and a price set in Chicago futures. When ethanol mandates raised corn prices in the late 2000s, farmers expanded acreage and farmland values and rents climbed — the above-normal returns were competed away into the price of land, just as long-run theory predicts.

## 2.5 Monopoly

*A single seller facing the whole demand curve.*

### Monopoly

A **monopoly** is the sole seller of a product without close substitutes, protected by **barriers to entry**: patents, control of a key resource, government licensing, or economies of scale so large that one firm serves the market most cheaply (a **natural monopoly**).

#### Monopoly pricing

Because the monopolist faces the downward-sloping market demand curve, selling another unit requires cutting the price on all units, so  $MR < P$ . It produces where  $MR = MC$  and charges the price buyers will pay for that quantity. The resulting markup obeys the **Lerner index**:

$$(P - MC) / P = -1 / E_d.$$

The less elastic the demand, the higher the markup.

### Welfare cost

A monopolist produces less than the competitive quantity and charges more, creating a **deadweight loss**. Policy responses include antitrust enforcement, and for natural monopolies, price regulation or public ownership.

### A real example

In 2015 Turing Pharmaceuticals acquired Daraprim, a decades-old drug with no competing generic in the U.S., and raised its price from \$13.50 to \$750 per pill. Demand from patients with a serious parasitic infection was extremely inelastic, and regulatory barriers kept rivals out — the Lerner formula at work.

## 2.6 Oligopoly and game theory

*When a few firms must anticipate each other's moves.*

### Oligopoly and game theory

An **oligopoly** has a few large firms whose decisions affect one another — airlines, wireless carriers, soft drinks. Because each firm's best choice depends on its rivals', oligopoly is analyzed with **game theory**.

### Nash equilibrium

A **Nash equilibrium** is a set of strategies in which no player can gain by changing its own strategy given the others':

$$u_i(s_i^*, s_{-i}^*) \geq u_i(s_i, s_{-i}^*) \text{ for every } s_i.$$

### The prisoner's dilemma

Firms would jointly earn the most by colluding like a monopoly, but each individually profits by cheating — cutting price or expanding output. When every firm follows its own interest, they end up at a worse outcome for all of them. This is why **cartels** are unstable and why collusion requires enforcement, repeated interaction, or explicit illegal agreements.

### A real example

OPEC assigns production quotas to its members, but each member gains by pumping above its quota while others restrain. In the early 1980s widespread cheating eroded prices, and in 1985–86 Saudi Arabia, tired of cutting its own output to compensate, flooded the market; oil prices collapsed by more than half. The cartel's history is a prisoner's dilemma played out in barrels.

## 2.7 Externalities and public goods

*When markets misprice costs and benefits that fall on others.*

### Externalities and public goods

An **externality** is a cost or benefit that falls on people outside a transaction. Pollution is a negative externality; vaccination is a positive one. With a negative externality, the social cost exceeds the private cost:

$$SMC = PMC + MEC,$$

## Correcting externalities

where MEC is the marginal external cost. The market ignores MEC and overproduces.

Remedies:

- A **Pigouvian tax** equal to MEC at the efficient quantity makes producers internalize the harm.
- **Cap-and-trade** fixes total emissions and lets firms trade permits, so cuts happen where they are cheapest.
- The **Coase theorem**: with clear property rights and low bargaining costs, private parties negotiate to the efficient outcome on their own.

## Public goods

A **public good** is *nonrival* (one person's use does not reduce another's) and *nonexcludable* (non-payers cannot be kept out) — national defense, basic research. Because everyone can **free-ride**, markets underprovide them, which is a core rationale for taxation.

## A real example

The U.S. Acid Rain Program, created by the 1990 Clean Air Act amendments, capped power-plant sulfur dioxide emissions and let utilities trade allowances. Emissions fell faster than required, and allowance prices ran far below early forecasts — utilities found cheap ways to cut pollution once pollution had a price.

## Week 3 — Macroeconomics: output, prices, and money

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Measuring GDP; inflation and price indices; unemployment; long-run economic growth; money and banking; aggregate demand and aggregate supply; business cycles.

### 3.1 Measuring GDP

*The market value of everything an economy produces.*

#### Measuring GDP

**Gross domestic product (GDP)** is the market value of all final goods and services produced within a country in a period. Because every dollar spent is a dollar of someone's income, GDP can be measured by spending, income, or value added — all three give the same total.

#### The expenditure identity

$$Y = C + I + G + NX,$$

where C is consumption, I investment (business equipment, structures, housing, inventories), G government purchases, and NX net exports (exports minus imports). Transfer payments such as Social Security are excluded because they buy no current output.

#### Nominal versus real

**Nominal GDP** uses current prices; **real GDP** uses constant base-year prices, so it rises only when output rises. Their ratio is the **GDP deflator**:

$$\text{Deflator} = (\text{Nominal GDP} / \text{Real GDP}) \times 100.$$

GDP omits household production, leisure, the underground economy, and environmental damage — it measures output, not well-being.

#### A real example

Consumption makes up roughly two-thirds of U.S. GDP, which is why consumer spending dominates economic news. The limits of the measure show in the old puzzle: if a man marries his housekeeper and she keeps doing the same work unpaid, GDP falls — though nothing real has changed.

### 3.2 Inflation and price indices

*Tracking the rising cost of living.*

#### Inflation and price indices

**Inflation** is a sustained rise in the general price level. It is measured with a price index; the most familiar is the **Consumer Price Index (CPI)**, which prices a fixed basket of goods a typical household buys.

#### The formulas

$$CPI_t = (\text{Cost of basket}_t / \text{Cost of basket}_{base}) \times 100,$$

$$\pi_t = (CPI_t - CPI_{t-1}) / CPI_{t-1}.$$

A fixed basket overstates inflation: it ignores **substitution** toward cheaper goods, **quality improvements**, and **new goods**. The 1996 Boskin Commission estimated the bias at about 1.1 percentage points a year.

### Costs of inflation

- **Shoe-leather costs** — effort spent holding less cash.
- **Menu costs** — the cost of changing prices.
- **Arbitrary redistribution** — unexpected inflation transfers wealth from lenders to borrowers.
- **Noise** — relative-price signals become harder to read.

### A real example

U.S. CPI inflation peaked at 9.1 percent in June 2022, the highest in four decades. At the extreme, Zimbabwe's hyperinflation in 2008 reached an estimated monthly rate of billions of percent, and the government printed a 100-trillion-dollar note. Prices doubled faster than workers could spend their wages, and the currency was abandoned.

## 3.3 Unemployment

*Who counts as unemployed, and why it never reaches zero.*

### Unemployment

A person is **unemployed** if they have no job but are actively looking for one. People not looking are outside the **labor force** entirely.

### The measures

$$u = U / (U + E) \times 100, LFPR = (U + E) / \text{Adult population} \times 100.$$

Discouraged workers who stop searching drop out of both the numerator and the denominator, so the official rate can fall even when the job market is weak.

### Types of unemployment

- **Frictional** — time spent matching workers to jobs.
- **Structural** — mismatch of skills or location with available jobs.
- **Cyclical** — the shortfall caused by recessions.

The **natural rate**  $u_n$  is frictional plus structural unemployment. **Okun's law** links the cyclical part to lost output:

$$(Y - Y^*) / Y^* \approx -2 (u - u_n).$$

### A real example

U.S. unemployment was 3.5 percent in February 2020 and 14.7 percent in April 2020 — the fastest rise on record, almost entirely cyclical as businesses closed. It fell back below 4 percent by early 2022, far faster than after the 2008 recession, because most job losses were temporary layoffs rather than permanent separations.

### 3.4 Long-run economic growth

*Why some countries are rich and others poor.*

#### Long-run economic growth

Over long horizons, nothing matters more for living standards than the growth rate of real GDP per person. Small differences in growth compound into enormous differences in income.

#### The production function

Output depends on capital  $K$ , labor  $L$ , and productivity  $A$ :

$$Y = A K^{\alpha} L^{1-\alpha}.$$

Taking growth rates gives **growth accounting**:

$$g_Y = g_A + \alpha g_K + (1 - \alpha) g_L.$$

Capital alone hits diminishing returns (the **Solow model**), so sustained growth in income per person must come from rising productivity — technology, education, and institutions.

#### The rule of 70

A quantity growing at  $g$  percent per year doubles in about

$$\text{Doubling time} \approx 70 / g \text{ years.}$$

#### A real example

In 1960 South Korea and Ghana had similar incomes per person. Korea then grew at roughly 6 percent per year for decades — doubling income about every 12 years — through heavy investment, mass education, and export-driven technology adoption. Today Korea's income per person is many times Ghana's. Growth, not redistribution, created the gap.

### 3.5 Money and banking

*What money is, how banks create it, and how it relates to prices.*

#### Money and banking

**Money** is any asset widely accepted in exchange. It serves three functions: a **medium of exchange**, a **unit of account**, and a **store of value**.

#### Fractional-reserve banking

Banks keep only a fraction  $rr$  of deposits as reserves and lend the rest; the loans are redeposited and lent again. In the simple model each dollar of reserves supports

$$m = 1 / rr$$

dollars of deposits — the **money multiplier**. In practice, excess reserves and cash holdings make the multiplier far smaller and unstable.

#### The quantity theory

$$MV = PY,$$

where  $M$  is the money supply,  $V$  velocity,  $P$  the price level, and  $Y$  real output. In growth rates,  $\pi \approx g_M + g_V - g_Y$ : if velocity is stable, sustained money growth beyond output growth

produces inflation.

### A real example

The U.S. M2 money supply grew by about 25 percent between early 2020 and early 2021, the fastest since World War II, and inflation followed in 2021–22. But velocity fell sharply during the pandemic, absorbing much of the new money — which is why inflation, though high, was far smaller than the money growth alone would predict.

## 3.6 Aggregate demand and aggregate supply

*The workhorse model of output and the price level.*

### Aggregate demand and aggregate supply

The **AD-AS model** determines real output and the price level together. It is supply and demand for the whole economy.

#### The curves

- **Aggregate demand** — total spending,  $Y = C + I + G + NX$ , slopes down: a lower price level raises real wealth, lowers interest rates, and makes exports cheaper.
- **Short-run aggregate supply** slopes up because wages and some prices are sticky.
- **Long-run aggregate supply** is vertical at potential output  $Y^*$ , set by resources and technology, not the price level.

#### Shocks

A **demand shock** moves output and prices in the same direction. A **supply shock** moves them in opposite directions — an adverse one raises prices while cutting output. In the long run, wages adjust and the economy returns to  $Y^*$ .

### A real example

The 1973 OPEC embargo quadrupled oil prices — a textbook adverse supply shock. The U.S. got **stagflation**: inflation above 10 percent and, by 1975, unemployment near 9 percent. Policymakers faced an impossible choice, since stimulating demand would worsen inflation and fighting inflation would deepen the slump.

## 3.7 Business cycles

*The recurring rise and fall of economic activity.*

### Business cycles

Real GDP grows along a long-run trend but fluctuates around it in **business cycles**: expansion, peak, contraction (recession), and trough. In the U.S., cycle dates are set by the National Bureau of Economic Research.

#### The output gap

$$\text{Output gap} = (Y - Y^*) / Y^* \times 100.$$

A negative gap means idle resources and rising unemployment; a positive gap means the economy is running hot and inflation tends to rise.

## Why cycles happen

- **Keynesian view** — swings in aggregate demand plus sticky prices.
- **Real business cycle view** — shocks to productivity and technology.
- **Financial accelerator** — credit booms and busts amplify small shocks.

Most modern models combine all three.

## A real example

The NBER dates the Great Recession from December 2007 to June 2009. U.S. real GDP fell about 4 percent from peak to trough, and the output gap stayed negative for years afterward because the collapse of housing credit — a financial accelerator — kept demand depressed long after the recession officially ended.

## Week 4 — Macroeconomic policy and the open economy

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Fiscal policy and the multiplier; monetary policy and central banking; the Phillips curve and expectations; exchange rates and purchasing power parity; the balance of payments; international trade policy; government debt and deficits; a capstone synthesis.

### 4.1 Fiscal policy and the multiplier

*Government spending and taxes as tools for stabilizing output.*

#### Fiscal policy and the multiplier

**Fiscal policy** is the use of government purchases, transfers, and taxes to influence aggregate demand. Expansionary policy raises spending or cuts taxes; contractionary policy does the reverse.

#### The multiplier

An extra dollar of government spending becomes someone's income; they spend a fraction — the **marginal propensity to consume** (MPC) — and so on. The rounds sum to

$$k = 1 / (1 - MPC), k_T = -MPC / (1 - MPC).$$

The tax multiplier is smaller because part of a tax cut is saved before any spending occurs.

#### Why real multipliers are smaller

- **Leakages** — taxes and imports absorb spending.
- **Crowding out** — government borrowing raises interest rates and reduces private investment.
- **Expectations** — households anticipating future taxes may save today.

**Automatic stabilizers** — unemployment insurance, progressive taxes — expand the deficit in recessions without any new legislation.

#### A real example

With  $MPC = 0.8$  the textbook multiplier is 5. Empirical estimates are far lower — typically between about 0.5 and 2. The 2009 American Recovery and Reinvestment Act, roughly \$800 billion, was evaluated by the Congressional Budget Office with multipliers in roughly that range depending on the component: direct purchases scored high, tax cuts for high earners scored low.

### 4.2 Monetary policy and central banking

*How the central bank steers interest rates, credit, and inflation.*

#### Monetary policy and central banking

A **central bank** — the Federal Reserve in the U.S. — conducts **monetary policy** by setting a short-term policy rate (the federal funds rate), buying and selling securities (open-market operations), and, when rates hit zero, large-scale asset purchases (**quantitative easing**).

## The Taylor rule

A widely used description of how central banks set rates:

$$i = r^* + \pi + 0.5 (\pi - \pi^*) + 0.5 (Y - Y^*)/Y^*,$$

where  $r^*$  is the neutral real rate and  $\pi^*$  the inflation target. The rule raises the nominal rate more than one-for-one with inflation, so the *real* rate rises and cools demand.

## Transmission

Higher policy rates raise borrowing costs, lower asset prices, strengthen the currency, and reduce spending — with **long and variable lags** of a year or more. Central-bank **credibility** matters because expectations of future policy drive long-term rates and wage setting.

## A real example

In 1979 Fed Chair Paul Volcker set out to break inflation running above 13 percent. The federal funds rate reached about 20 percent in 1981; the economy entered a deep recession, and unemployment hit 10.8 percent in late 1982. By 1983 inflation had fallen to roughly 3 percent, and the Fed's credibility has anchored expectations ever since.

## 4.3 The Phillips curve and expectations

*The short-run tradeoff between inflation and unemployment — and why it vanishes.*

### The Phillips curve and expectations

In 1958 A. W. Phillips found that years of low unemployment in Britain were years of rapidly rising wages. The **Phillips curve** seemed to offer policymakers a menu: accept more inflation, get less unemployment.

### The expectations-augmented curve

Milton Friedman and Edmund Phelps argued the tradeoff exists only relative to **expected inflation**:

$$\pi = \pi^e - \beta (u - u_n).$$

Unemployment falls below the natural rate only when inflation *surprises* workers. Once expectations catch up, unemployment returns to  $u_n$  at a higher inflation rate: the **long-run Phillips curve is vertical**.

## Implications

Policy cannot permanently buy lower unemployment with inflation. Anchoring expectations becomes central, which is why modern central banks announce explicit inflation targets.

## A real example

The 1960s U.S. data fit Phillips's tradeoff neatly. Policymakers leaned on it, inflation expectations rose, and the 1970s delivered high inflation *and* high unemployment — the curve shifted exactly as Friedman had predicted in 1967. In the 2010s the opposite puzzle appeared: unemployment fell to historic lows with little inflation, because expectations were firmly anchored near 2 percent.

## 4.4 Exchange rates and purchasing power parity

*The price of one currency in terms of another.*

### Exchange rates and purchasing power parity

The **nominal exchange rate**  $e$  is the number of units of foreign currency one unit of domestic currency buys. What matters for trade is the **real exchange rate**, which compares the price of goods across countries:

$$\varepsilon = e \cdot P / P^*,$$

### Purchasing power parity

where  $P$  and  $P^*$  are domestic and foreign price levels. If goods can be freely arbitrated, the real exchange rate should equal 1 — **purchasing power parity (PPP)**:

$$e = P^* / P.$$

In growth rates, **relative PPP** says the currency of the higher-inflation country depreciates by the inflation differential:  $\% \Delta e \approx \pi^* - \pi$ .

### Why PPP fails in the short run

Many goods and services are not tradable, transport costs and tariffs block arbitrage, and capital flows move currencies far more than trade does. PPP holds reasonably well over decades and poorly over months.

### A real example

The *Economist*'s **Big Mac index** compares the price of the same burger across countries, converted at market exchange rates, as a rough test of PPP. It routinely shows large, persistent gaps — Swiss burgers far dearer than Indian ones — because labor, rent, and other nontradable inputs make up most of the burger's cost.

## 4.5 The balance of payments

*The accounting of a nation's transactions with the rest of the world.*

### The balance of payments

The **balance of payments** records all transactions between a country and the rest of the world. The **current account** covers trade in goods and services plus income flows; the **financial account** covers purchases and sales of assets. Every transaction has two sides, so the accounts sum to zero.

### The saving-investment identity

From  $Y = C + I + G + NX$ , national saving  $S = Y - C - G$ , so

$$NX = S - I.$$

A country that invests more than it saves must import the difference from abroad: a current-account deficit is simply a financial-account surplus — foreigners acquiring domestic assets.

## Interpreting deficits

A current-account deficit is not inherently bad: a fast-growing economy with good investment opportunities borrows rationally. It becomes dangerous when it finances consumption rather than productive investment, or when it depends on short-term foreign funding that can flee.

### A real example

The U.S. has run current-account deficits almost every year since the early 1980s, financed largely by foreign purchases of Treasury bonds and other U.S. assets. China's current-account surplus reached nearly 10 percent of GDP in 2007 — the mirror image: very high Chinese saving flowing abroad, much of it into U.S. assets.

## 4.6 International trade policy

*Tariffs, quotas, and who pays for protection.*

### International trade policy

Governments restrict trade with **tariffs** (taxes on imports), **quotas** (limits on quantity), and subsidies to domestic producers. The case for free trade rests on comparative advantage; the case for protection rests on jobs, national security, infant industries, and bargaining leverage.

#### The effects of a tariff

A per-unit tariff  $t$  raises the domestic price by up to  $t$ . Domestic producers gain, consumers lose, and the government collects

$$\text{Tariff revenue} = t \times M,$$

where  $M$  is the quantity imported. The net loss to the country is two deadweight triangles — overproduction by inefficient domestic firms and underconsumption by buyers:

$$DWL = \frac{1}{2} t (\Delta Q_s + \Delta Q_d).$$

#### Political economy

The gains from protection are concentrated on a few producers who lobby hard; the costs are spread thinly across millions of consumers who rarely notice. That asymmetry explains why tariffs persist despite their net cost.

### A real example

In 2009 the U.S. imposed tariffs on Chinese tires. The Peterson Institute estimated the tariff saved about 1,200 jobs at a cost to American consumers of about \$1.1 billion — roughly \$900,000 per job saved — while China retaliated against U.S. chicken exports. Concentrated benefits, diffuse costs.

## 4.7 Government debt and deficits

*How public debt evolves and when it becomes a problem.*

### Government debt and deficits

The **budget deficit** is the excess of government spending (including transfers and interest) over revenue in a year; the **debt** is the accumulated stock of past deficits. What matters for sustainability is debt relative to the economy, the **debt-to-GDP ratio**  $b$ .

### Debt dynamics

The ratio evolves according to

$$\Delta b \approx (r - g) b - s,$$

where  $r$  is the interest rate on debt,  $g$  the growth rate of GDP, and  $s$  the primary surplus (revenue minus non-interest spending) as a share of GDP. When  $r < g$ , a country can run modest primary deficits while its debt ratio stays stable or falls; when  $r > g$ , it must run primary surpluses just to stand still.

### Costs of high debt

Heavy borrowing can crowd out private investment, raise future taxes, and, if lenders lose confidence, trigger a spiral of rising interest rates. Countries that borrow in their own currency have more room than those borrowing in foreign currency.

### A real example

Britain ended World War II with debt above 200 percent of GDP. It never repaid most of it; over the next three decades growth and inflation, with interest rates held low, shrank the ratio dramatically —  $r$  below  $g$  did the work. Japan today carries gross debt well above 200 percent of GDP with low interest costs for the same reason.

## 4.8 Capstone synthesis

*One principle behind everything: optimizing agents meeting in equilibrium.*

### Capstone synthesis

Step back, and the entire course reduces to a single sentence: **economic outcomes are equilibria reached by agents who choose at the margin under scarcity.**

### The one condition

$$MB = MC.$$

Everything you have learned applies that condition to a different decision-maker:

- **Consumers** equalize marginal utility per dollar (Week 2).
- **Firms** set marginal revenue equal to marginal cost (Week 2).
- **Markets** aggregate those choices into supply, demand, and equilibrium (Week 1).
- **Economies** aggregate markets into output, prices, and employment (Week 3).
- **Policymakers** weigh marginal benefits and costs of intervention, subject to expectations and open-economy constraints (Week 4).

## **Where markets fail**

The condition delivers efficiency only when private marginal benefits and costs equal social ones. Externalities, market power, public goods, and sticky prices break that link — and each break is exactly where policy has a role.

## **A real example**

Analyzing a carbon tax uses the whole course at once: externality theory sets the tax at the marginal damage (Week 2); incidence analysis shows who actually pays (Week 1); macroeconomics traces its effect on prices, output, and the fiscal accounts (Weeks 3–4); and open-economy analysis asks whether production simply moves abroad, motivating border adjustments (Week 4). Master the marginal logic and you can analyze any policy.

# Practice problems

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A sample of the graded work in the course: two homeworks plus the midterm and final. Answers follow at the end.

## Homework 1.1 — Supply, demand, and equilibrium

*Short-answer problems on markets. Write each answer as the key formula in symbols using the on-screen math keyboard (use the Economics and Statistics tabs for Greek letters and  $\Delta$ ).*

1. Write the linear demand function with intercept  $a$  and slope  $b$ .
2. Write the linear supply function with intercept  $c$  and slope  $d$ .
3. Write the market equilibrium condition at price  $P^*$ .
4. Using the linear curves above, write the equilibrium price  $P^*$ .

## Homework 1.2 — Elasticity, controls, and taxes

*Use the math keyboard for fractions, subscripts,  $\Delta$ , and the percent sign.*

1. Write the price elasticity of demand in terms of percentage changes.
2. Write the midpoint formula for elasticity between  $(P_1, Q_1)$  and  $(P_2, Q_2)$ .
3. Write the tax wedge linking buyer price  $P_b$ , seller price  $P_s$ , and per-unit tax  $t$ .
4. Write the deadweight loss of a per-unit tax  $t$  that reduces quantity by  $\Delta Q$ .

## Midterm — Weeks 1 & 2

*Cumulative midterm on markets and on consumers, firms, and market structure. 60 minutes. Math keyboard available; pasting disabled.*

1. Write the opportunity cost of  $X$  along a PPF in terms of  $\Delta X$  and  $\Delta Y$ .
2. Write the price elasticity of demand.
3. Write the equilibrium price for  $Q_d = a - bP$  and  $Q_s = c + dP$ .
4. Write the budget constraint for goods  $X$  and  $Y$  with prices  $P_x$ ,  $P_y$  and income  $I$ .
5. Write the consumer's optimality condition in terms of marginal utilities and prices.
6. Write average total cost (ATC).
7. Write the profit-maximizing condition for a perfectly competitive firm.
8. Write the Lerner index for a monopolist.

## Final Exam — Economics

*Cumulative final covering all four weeks. 90 minutes. Math keyboard available; pasting disabled.*

1. Write the price elasticity of demand.
2. Write consumer surplus for linear demand with choke price  $P_{\max}$ , price  $P^*$ , and quantity  $Q^*$ .
3. Write the Lerner index for a monopolist.
4. Write the GDP expenditure identity.
5. Write the unemployment rate  $u$  in terms of unemployed  $U$  and employed  $E$ .
6. Write the quantity equation of money.
7. Write the Cobb-Douglas production function with productivity  $A$  and capital share  $\alpha$ .
8. Write the government-spending multiplier  $k$  in terms of the MPC.
9. Write the Taylor rule with neutral rate  $r^*$ , inflation  $\pi$ , target  $\pi^*$ , and output gap.
10. Write the expectations-augmented Phillips curve.

## Answer key

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### Homework 1.1 — Supply, demand, and equilibrium:

1.  $Q_d = a - bP$
2.  $Q_s = c + dP$
3.  $Q_d(P^*) = Q_s(P^*)$
4.  $P^* = (a - c) / (b + d)$

### Homework 1.2 — Elasticity, controls, and taxes:

1.  $E_d = \% \Delta Q_d / \% \Delta P$
2.  $E = [(Q_2 - Q_1) / ((Q_1 + Q_2)/2)] / [(P_2 - P_1) / ((P_1 + P_2)/2)]$
3.  $P_b - P_s = t$
4.  $DWL = \frac{1}{2} \cdot t \cdot \Delta Q$

### Midterm — Weeks 1 & 2:

1.  $OC_X = -\Delta Y / \Delta X$
2.  $E_d = \% \Delta Q_d / \% \Delta P$
3.  $P^* = (a - c) / (b + d)$
4.  $P_x X + P_y Y = I$
5.  $MU_x / P_x = MU_y / P_y$
6.  $ATC = TC / Q$
7.  $P = MR = MC$
8.  $(P - MC) / P = -1 / E_d$

**Final Exam — Economics:**

1.  $E_d = \% \Delta Q_d / \% \Delta P$
2.  $CS = \frac{1}{2} (P_{\max} - P^*) Q^*$
3.  $(P - MC) / P = -1 / E_d$
4.  $Y = C + I + G + NX$
5.  $u = U / (U + E) \times 100$
6.  $MV = PY$
7.  $Y = A K^\alpha L^{(1-\alpha)}$
8.  $k = 1 / (1 - MPC)$
9.  $i = r^* + \pi + 0.5(\pi - \pi^*) + 0.5(Y - Y^*)/Y^*$
10.  $\pi = \pi^e - \beta(u - u_n)$